IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Jennifer Brink

22-04-2026

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Data was collected through a SpaceX API as well as through data scraping, data wrangling was then used to make the data useable for our purposes.
- Exploratory data analysis was performed using visualization techniques such as scatter plots, bar charts as well as line plots. Additional analysis was performed using SQL queries.
- Interactive visual analytics were created using Folium (to visualize launch sites) and an interactive dashboard was created with Plotly Dash.
- Finally, predictive analysis was performed using classification models such as Logistic Regression, SVM, Decision Tree and KNN, with the Decision Tree model performing the best according to the accuracy scores.
- The results proved that there is a relation between the number of successes with the number of flights and orbit types.
- The number of successes increased over time and all launch sites are within close proximities to coastlines, railways and highways while being a decent distance from major cities to avoid disturbances.

Introduction

In this project we want to predict whether the first stage of a Falcon 9 rocket from SpaceX will land successfully. If the landing is successful SpaceX claims, it can reduce the rocket cost down from the industry standard of 165 million dollars to 62 million dollars.

This reduce in cost is due to the fact that if the first stage can land successfully it is reusable.

Thus, the main aim of this project is to be able to answer the question of whether the first stage of a Falcon 9 rocket can be reused based on factors such as orbit type and the launch site used.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - Data was collected using a SpaceX API as well as through data scraping.
- Perform data wrangling
  - Data was processed with data wrangling by mainly calculating the total number of successful landings for certain attributes such as orbit types and launch sites.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Logistic Regression, SVM , KNN and Decisions tree classification models were created and trained using training data, and then their performance was analyzed based on accuracy scores by use of testing data.

Data Collection

Data was collected through SpaceX API as well as data scraping methods. Additionally, the data was cleaned up and made useable through data wrangling processes.

Data Collection – SpaceX API

1. Request the SpaceX data using a GET request.

2. Parse the data and convert it into a data frame using Pandas.

3. Extract only the useful columns.

4. Filter the data frame to contain only Falcon 9 data.

5. Handel missing data values for Payload Mass by using the mean value to replace missing values.

6. Export to a CSV file.

[GitHub link]

Data Collection - Scraping

1. Request the SpaceX data using a HTTP GET request.

2. Create a BeautifulSoup object using the HTML response.

3. Extract only the relevant column names from the HTML table header.

4. Parse the data and convert it into a data frame using Pandas.

5. Export to a CSV file.

Data Scraping

Data Wrangling

1. Create a data frame using pd.read_csv from Pandas.

2. Calculate the number of launches from each of the launch sites.

3. Calculate the number and occurrences of each orbit type.

4. Calculate the number and occurrences of each mission outcome of the orbits.

5.Create a landing class column to indicate whether the landing was successful or not using binary variables..

6. Export to a CSV file.

[Data Wrangling]()

EDA with Data Visualization

- Scatter plots: used to compare or determine the relationship between two variables. In this analysis it was used to determine the relationships between:
  - Flight Number vs Payload Mass
  - Flight Number vs Launch Site
  - Payload Mass vs Launch Site
  - Flight Number vs Launch Site
  - Payload Mass vs Orbit Type
- Bar Charts: were used to visualize relationships between variables namely Orbit Type vs Success Rate.
- Line Charts: were used to analyze trend over time specifically the success rate over years.

Visualization notebook

EDA with SQL

Summary of SQL queries:

- Display the names of the unique launch sites in the space mission
- Display 5 records where launch sites begin with the string 'CCA'
- Display the total payload mass carried by boosters launched by NASA (CRS)
- Display average payload mass carried by booster version F9 v1.1
- List the date when the first successful landing outcome in ground pad was achieved.
- List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
- List the total number of successful and failure mission outcomes
- List all the booster versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.
- List the records which will display the month names, failure landing outcomes in drone ship ,booster versions, launch site for the months in year 2015.
- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

SQL notebook

Build an Interactive Map with Folium

Map objects used:

- Circles: used a radius of 1000 and used to mark the different launch sites.
- Cluster Markers: used to pinpoint the location of successful and unsuccessful landings.
- Mouse Position: used to easily add lines that are used to calculate the distances between launch sites and their nearest costliness, cities, railways and highways.

Folium notebook

Build a Dashboard with Plotly Dash

Plots and charts added to the dashboard:

- Pie chart of the launch success rates for all launch sites.
- Scatter plots showing the relationships between payload mass and the class.
- Dropdown menu where one can select to show the data of all sites or choose between the distinct launch sites and view only their data.
- The scatter plot changes based on the selection in the dropdown menu.
- The scatter plot also displays the booster versions based on which launch site is selected with the dropdown menu.
- A payload range slider to select which ranges payload mass one want to see data for.

Plotly dash code

Predictive Analysis (Classification)

1. Create a data frame using pd.read_csv from Pandas.

2. Create a Numpy array using the ‘Class’ column’s data.

3. Scale the data frame using StandardScaler.

4. Split the data into training and testing sets.

5.Use GridSearch to create objects for Logistic Regression, SVM, Decision Tree and KNN models. And find the best hyperparameters for each model using the train data.

6. Calculate accuracy and test scores.

7. Create confusion matrices for each model using the test data.

[Analysis notebook]

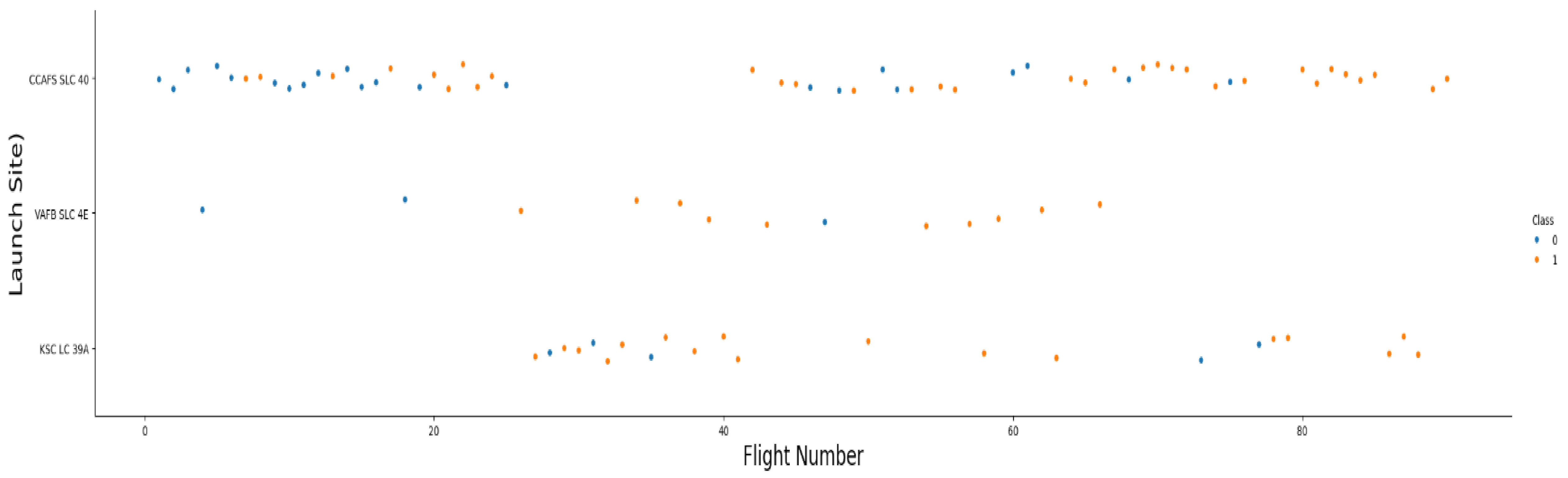
Results

- Exploratory data analysis results:
  revealed that launch success rates increased over time and that higher orbits had higher success rates.
- Interactive analytics demo in screenshots:
  revealed that there is a higher success rate for higher payload masses, booster versions v1.0 and v1.1 had lower success rates while versions FT, B4 and B5 had higher success rates. Launch site KSC LC-39A had the highest success rate while CCAFS SLC-40 had the lowest.
- Predictive analysis results:
  showed that the Decision Tree model had the best accuracy at 88.89% while all three the other models had an accuracy score of 83.33%.
- The overall success rate of Falcon 9 landings was 66.66%

Section 2

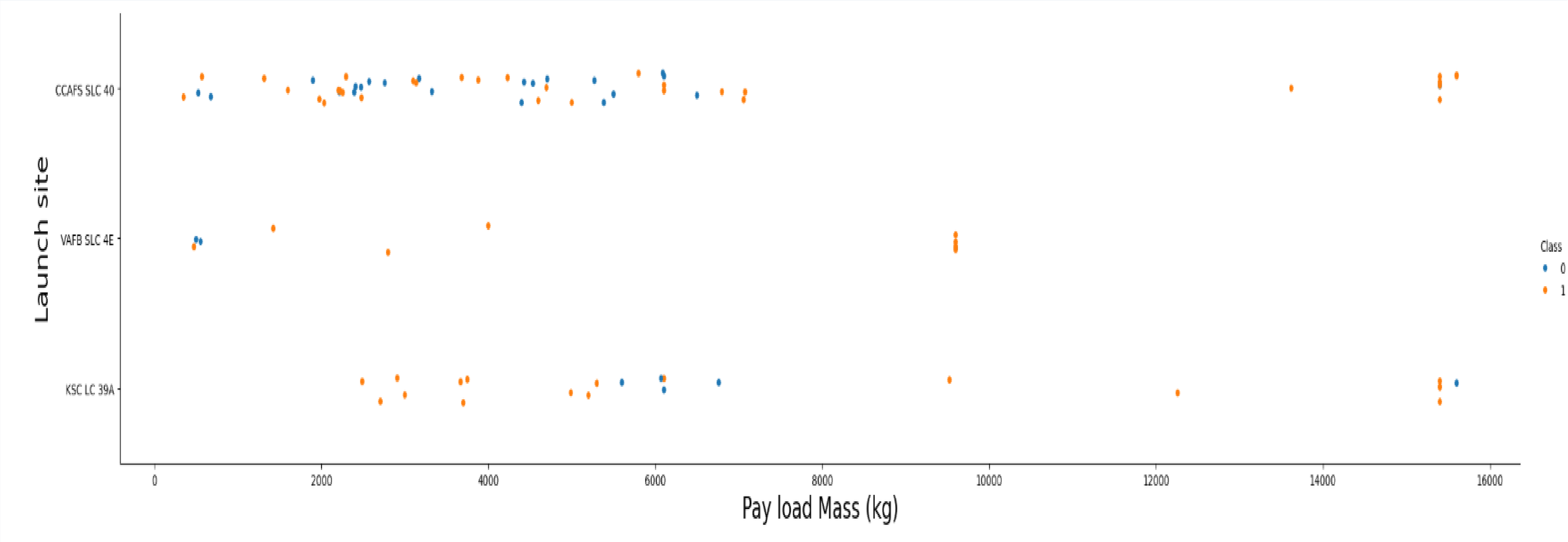
Insights drawn from EDA

Flight Number vs. Launch Site



- The number of successful launches increased as the number of flights increased.
- Class 0 (blue dots) represents failed landings, and class 1 (orange dots) represents successful landings.

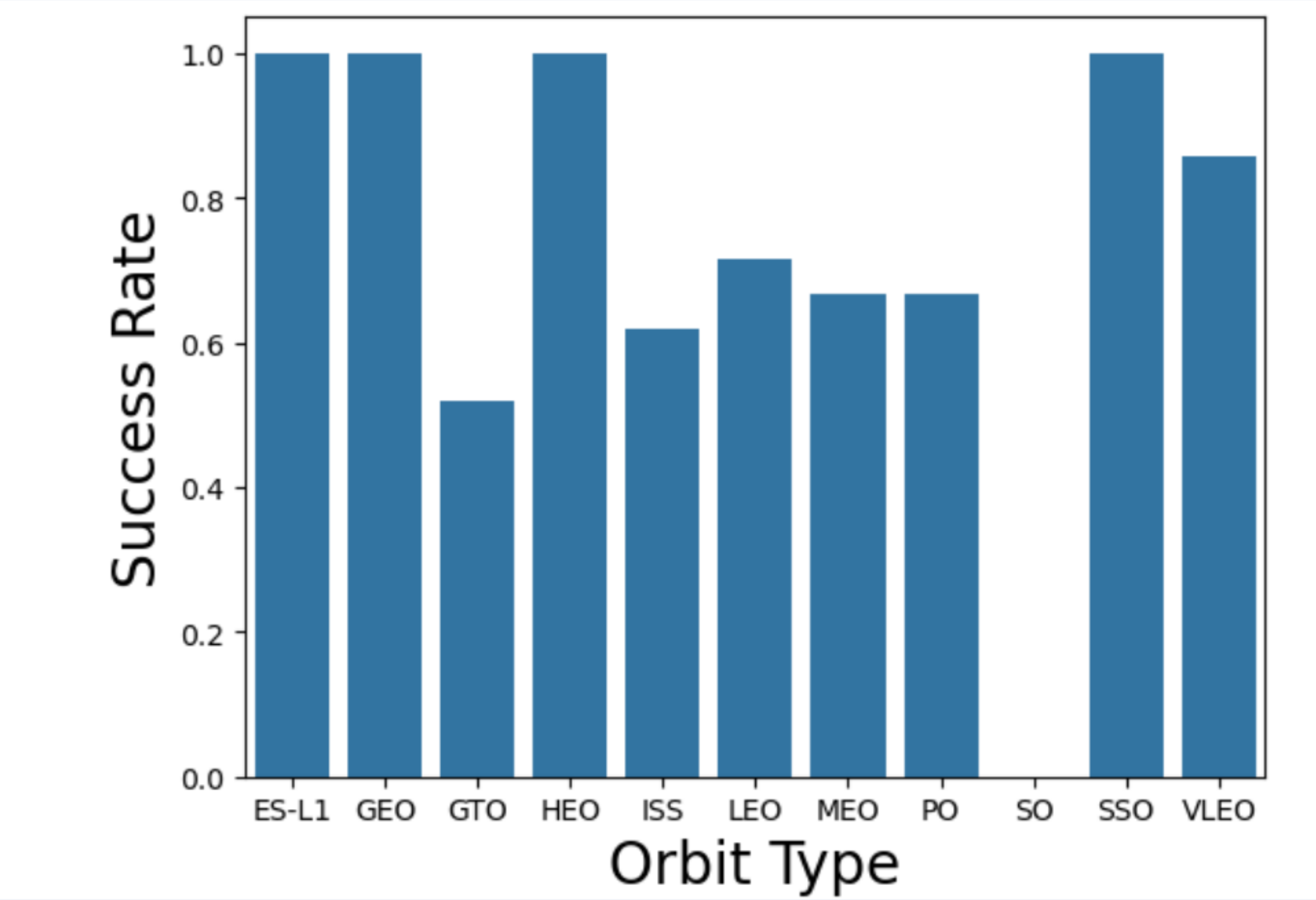
Payload vs. Launch Site



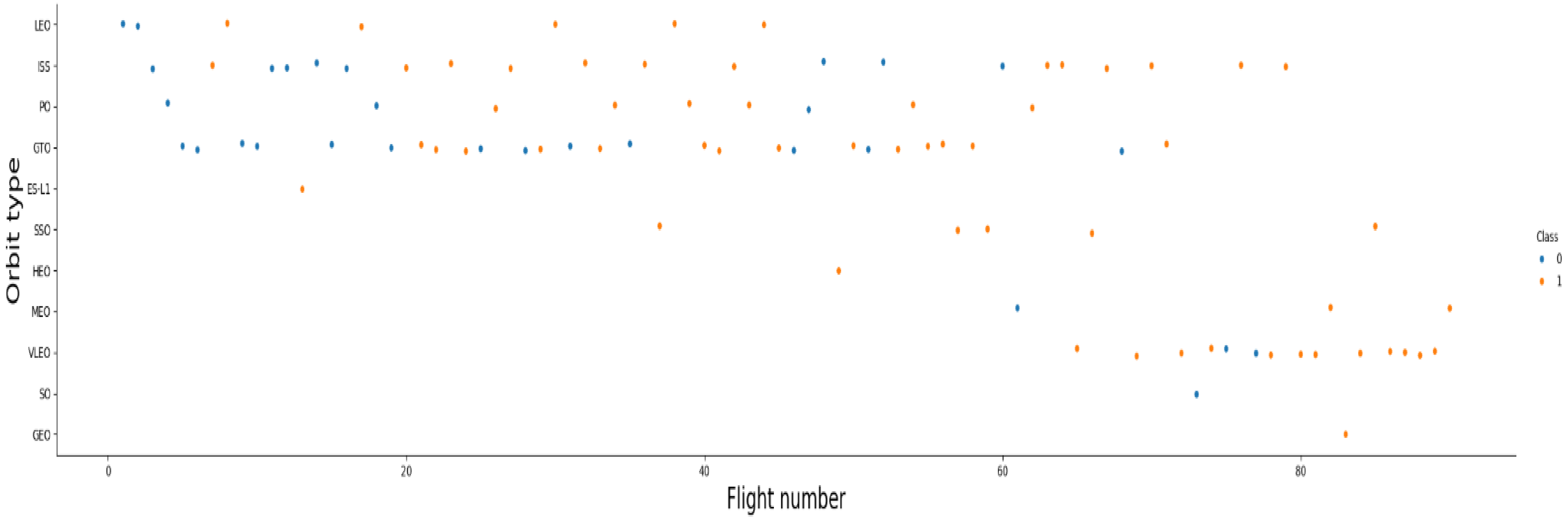
- There is a larger number of successful launches with a higher payload mass

Success Rate vs. Orbit Type

- ES-L1, GEO, HEO and SSO have the highest success rates, which are all high Earth orbits.
- The other orbits are all low Earth orbits. The SO orbit had no successes.

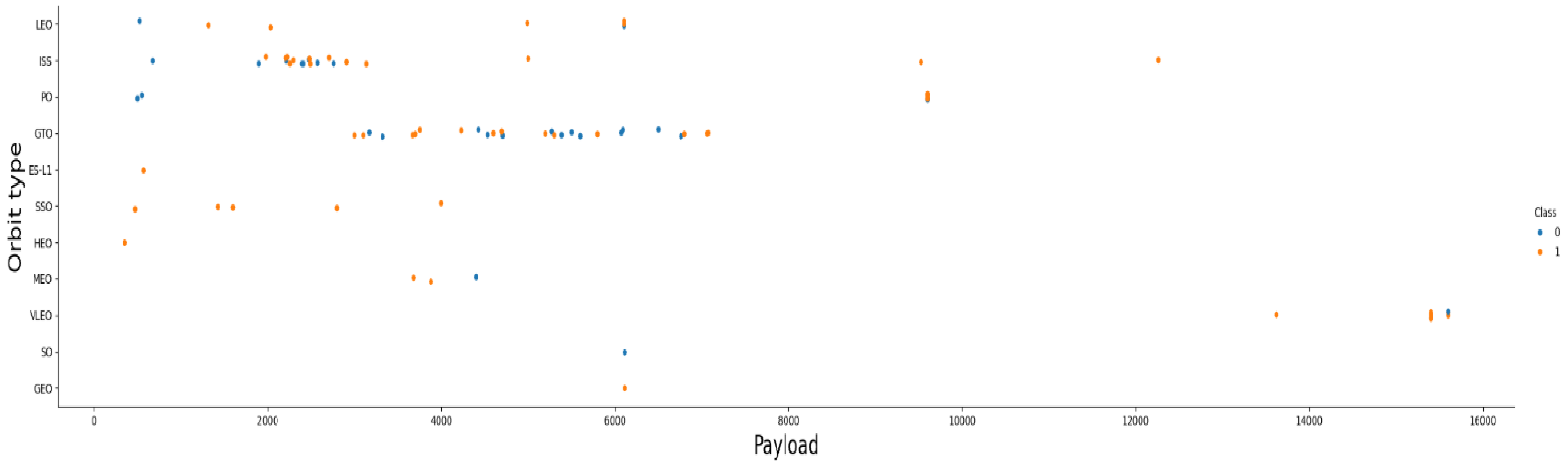


Flight Number vs. Orbit Type



- As the number of flights increases so does the number of successes.
- The number of successes aligns with that of the bar chart for each orbit type.

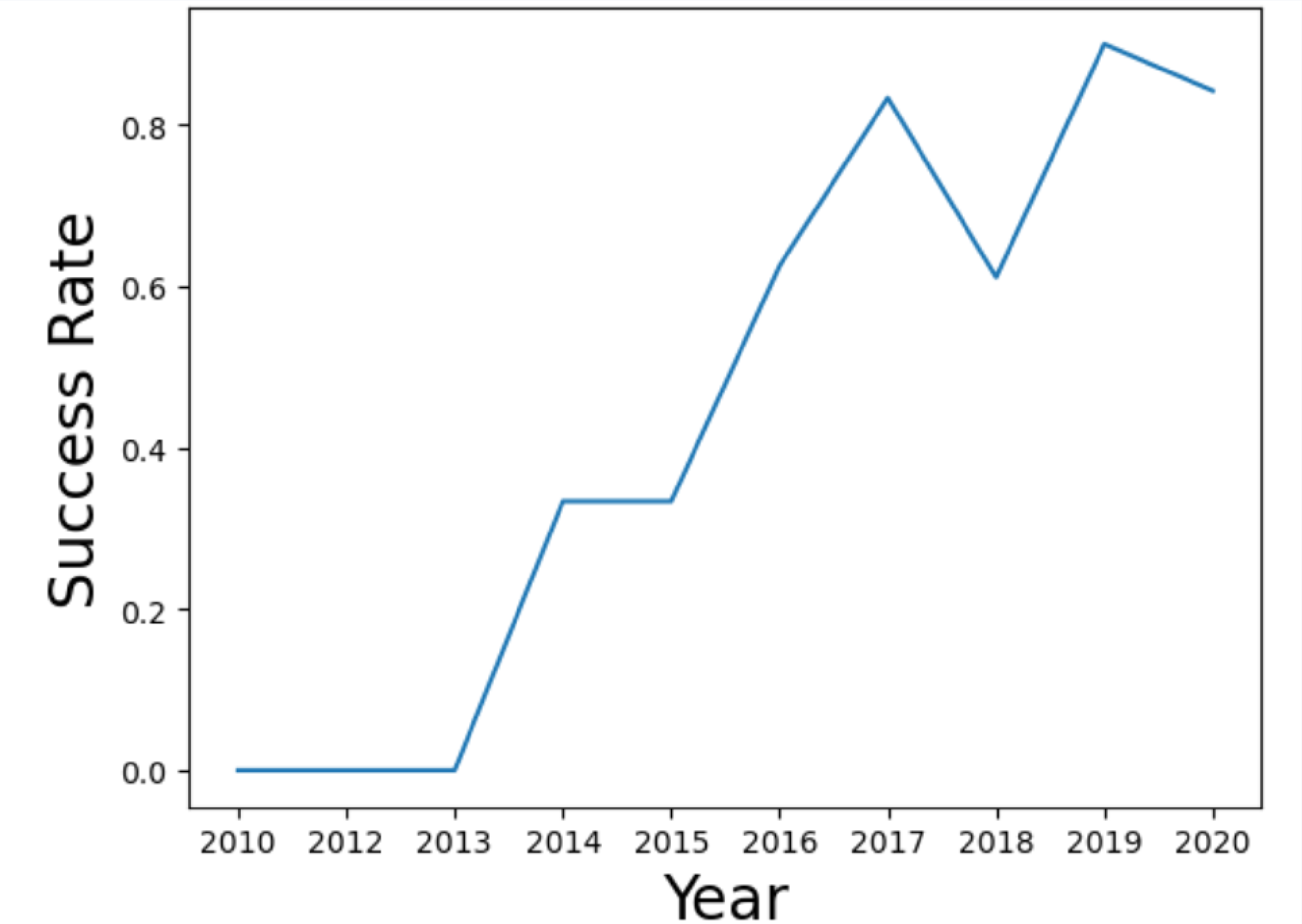
Payload vs. Orbit Type



- There seems to be no overall relation between Payload and orbit types when it comes to success rates.

Launch Success Yearly Trend

- As the years increase there are more successes.



All Launch Site Names

- Cape Canaveral Space Launch Complex 40 = CCAFS LC-40, CCAFS SLC-40
- Kennedy Space Center Launch Complex 39A = KSC LC-39A
- Vandenberg Space Launch Complex 4 = VAFB SLC-4E

	Launch_Site
0	CCAFS LC-40
1	VAFB SLC-4E
2	KSC LC-39A
3	CCAFS SLC-40

Launch Site Names Begin with 'CCA'

	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
0	2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
1	2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of...	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2	2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
3	2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
4	2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

5 sample records for launch sites beginning with ‘CCA’

Total Payload Mass

	sum(PAYLOAD_MASS__KG_)
0	45596

The total payload mass (in Kg) carried by boosters from NASA was calculated to be 45596.

Average Payload Mass by F9 v1.1

	avg(PAYLOAD_MASS__KG_)
0	2928.4

The average payload mass carried by booster version F9 v1.1 is given to be 2928.4.

First Successful Ground Landing Date

	min(Date)
0	2015-12-22

The first successful landing occurred on the 22nd of December 2010.

Successful Drone Ship Landing with Payload between 4000 and 6000

The boosters which have successfully landed with a payload mass between 4000 and 6000 are:

- F9 FT B1022
- F9 FT B1026
- F9 FT B1021.2
- F9 FT B1031.2

	Booster_Version
0	F9 FT B1022
1	F9 FT B1026
2	F9 FT B1021.2
3	F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

	Mission_Outcome	count(*)
0	Failure	1
1	Success	100

There was 100 successful missions while only 1 failure.

Boosters Carried Maximum Payload

The boosters which carried the maximum payload mass are:

- F9 B5 B1048.4
- F9 B5 B1049.4
- F9 B5 B1051.3
- F9 B5 B1056.4
- F9 B5 B1048.5
- F9 B5 B1051.4
- F9 B5 B1049.5
- F9 B5 B1060.2
- F9 B5 B1058.3
- F9 B5 B1051.6
- F9 B5 B1060.3
- F9 B5 B1049.7

	Booster_Version
0	F9 B5 B1048.4
1	F9 B5 B1049.4
2	F9 B5 B1051.3
3	F9 B5 B1056.4
4	F9 B5 B1048.5
5	F9 B5 B1051.4
6	F9 B5 B1049.5
7	F9 B5 B1060.2
8	F9 B5 B1058.3
9	F9 B5 B1051.6
10	F9 B5 B1060.3
11	F9 B5 B1049.7

2015 Launch Records

The failed landing outcomes in 2015 are:

	Month	Landing_Outcome	Booster_Version	Launch_Site
0	01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
1	04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

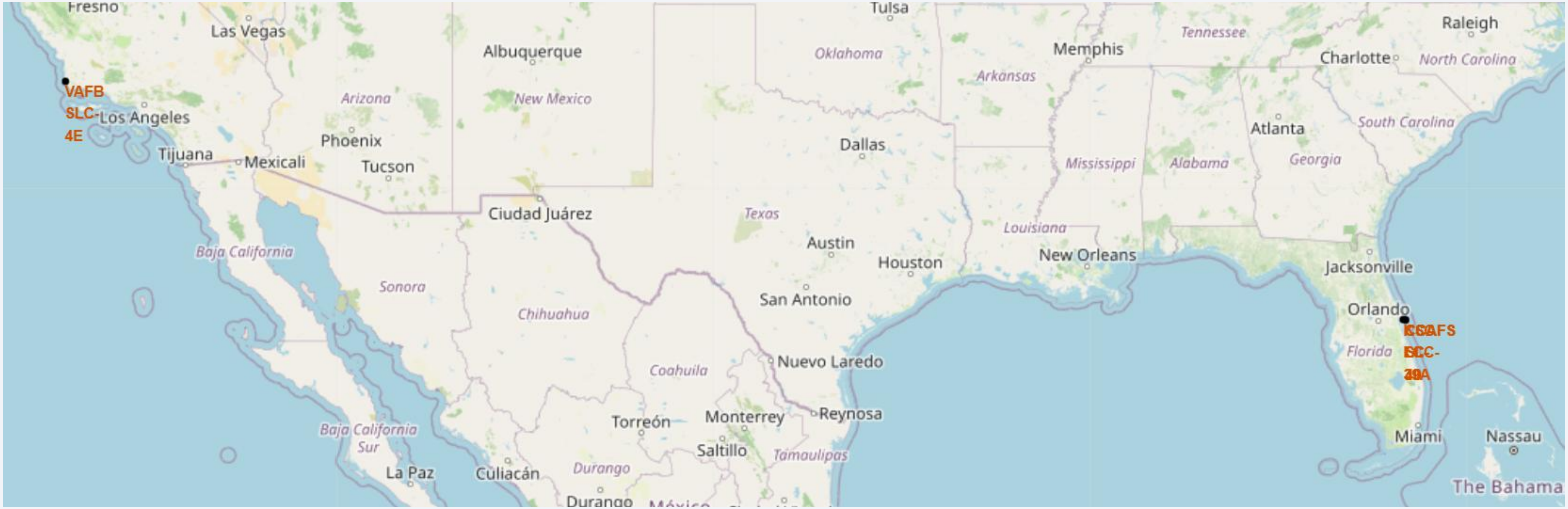
The landing outcomes for a certain time period are given by:

	Landing_Outcome	Count_Launches
0	No attempt	10
1	Success (drone ship)	5
2	Failure (drone ship)	5
3	Success (ground pad)	3
4	Controlled (ocean)	3
5	Uncontrolled (ocean)	2
6	Failure (parachute)	2
7	Precluded (drone ship)	1

Section 3

Launch Sites Proximities Analysis

Launch site Locations



The image shows the all the launch sites used by making use of circle markers, upon enlargement of the image it is visible that the launch sites do not overlap by are near each other.

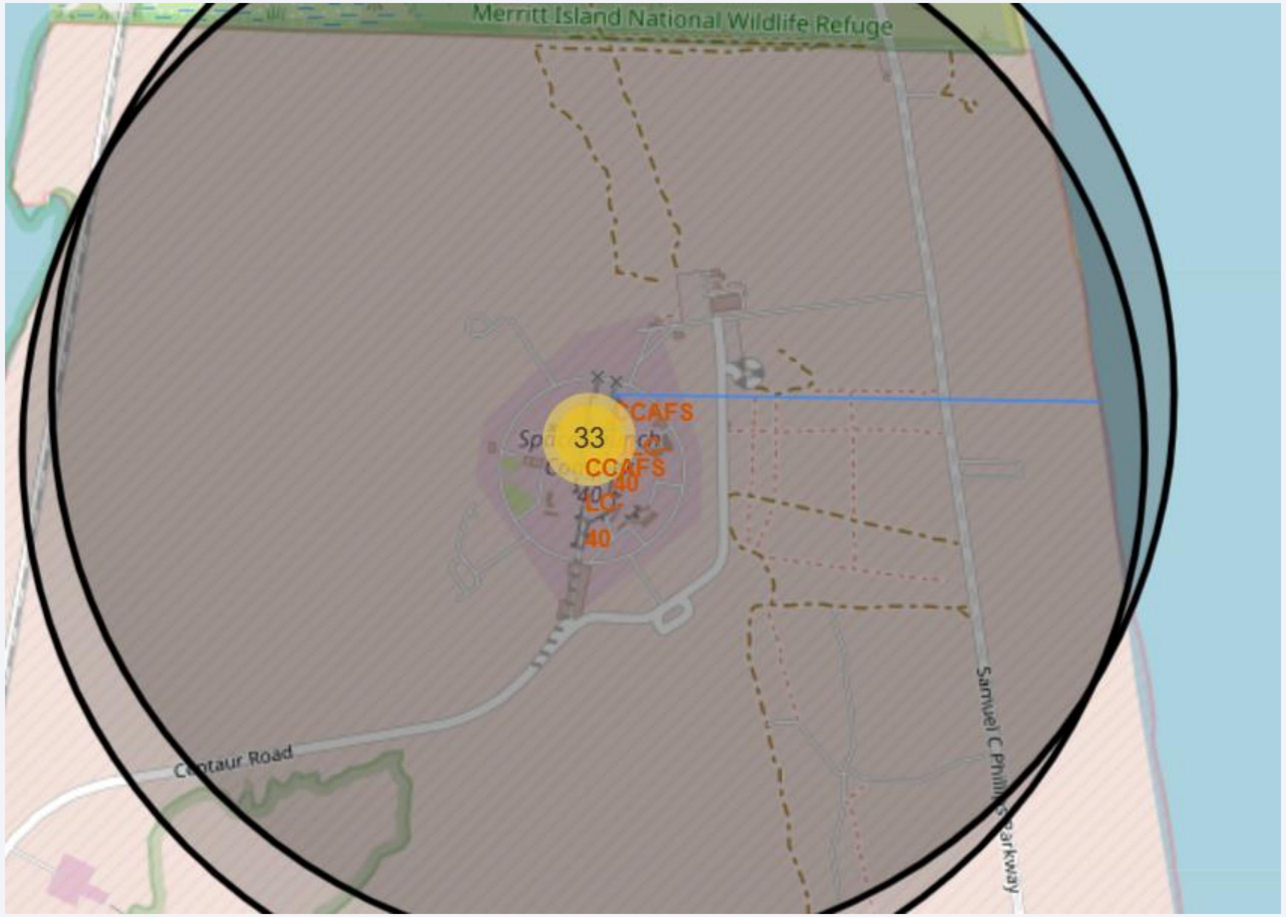
Colour coded launch outcomes

The image shows the number of successful (green) and unsuccessful (red) launches at a specific launch site.

Launch sites and their proximities

The launch sites have numerous proximities such as highways, railways, coastlines and cities.

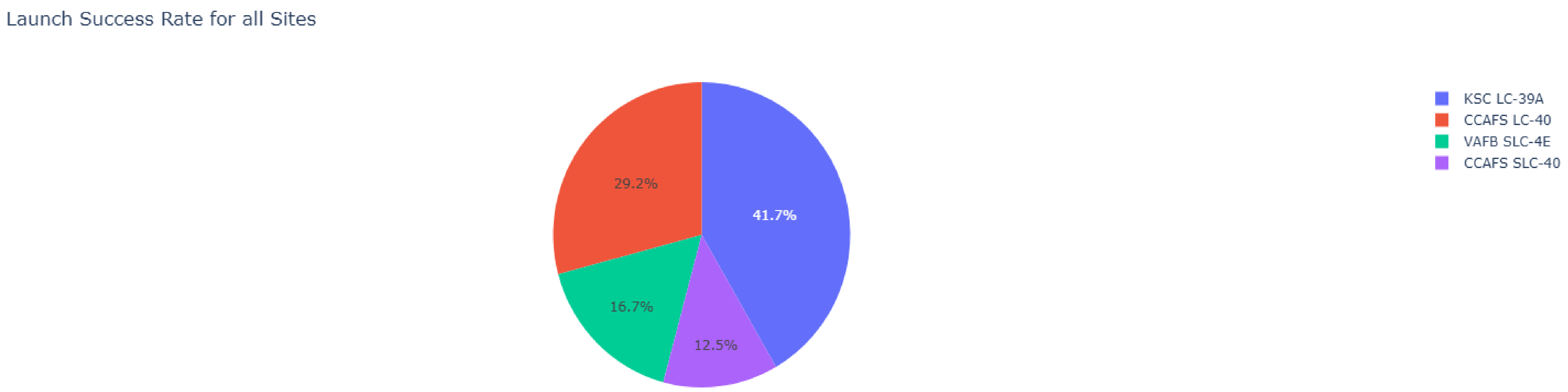
The distance between the launch site and its proximities are given by a blue line.



Section 4

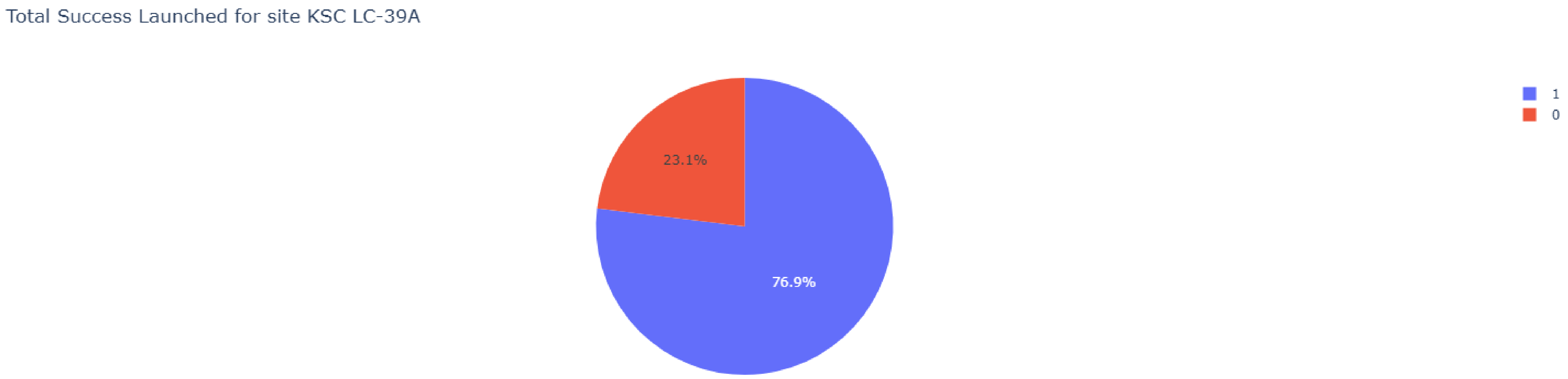
Build a Dashboard with Plotly Dash

Percentage of launch successes from each launch site



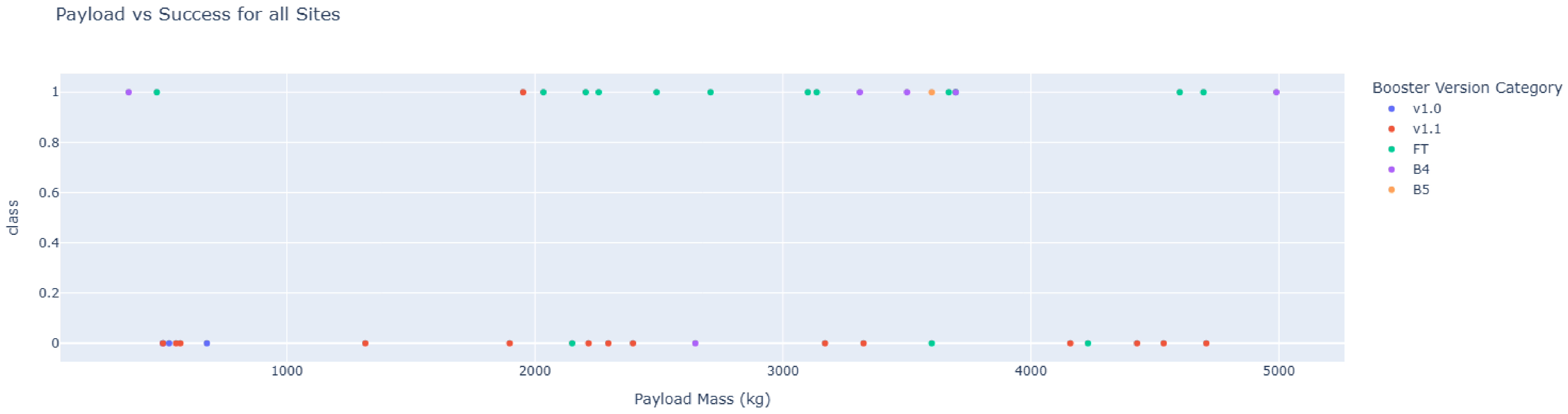
The KSC launch site had the highest number of successful landings with 41.7%, while the CCAFS site had the lowest with 12.5%.

The percentage of successful launches from the KSC site



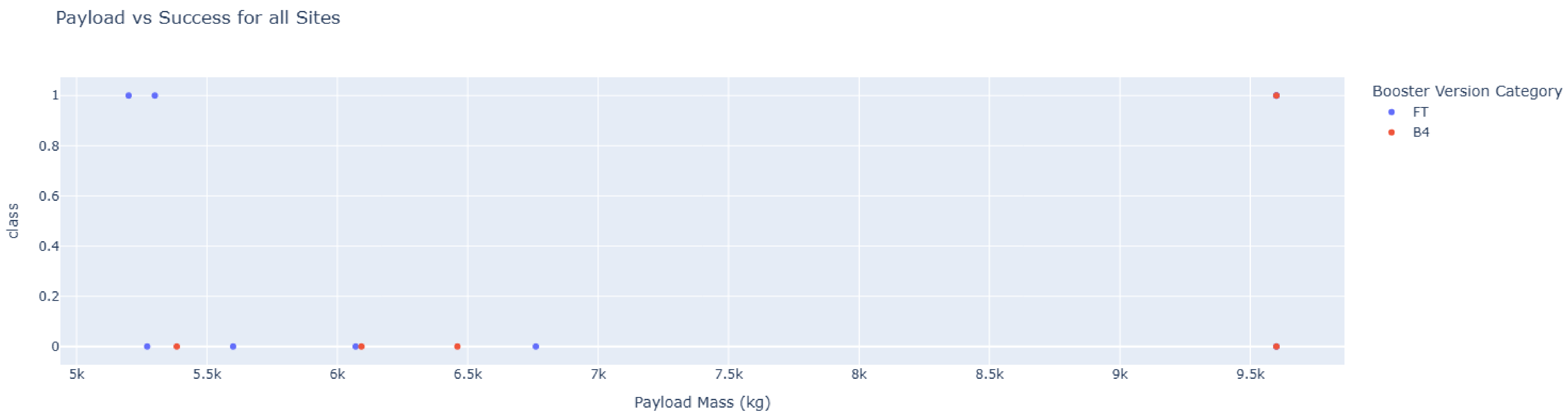
The KSC launch site had a ratio of 76.9% successful landings while 23.1% were unsuccessful.

Payload vs Launch outcome for all sites with a payload range up to 5000 Kg



All booster versions feature in this payload range. Booster versions v1.0, v1.1 and B5 mostly had unsuccessful launches, while booster versions FT, B4 had mostly successful launches.

Payload vs Launch outcome for all sites with a payload range of 5000 to 10000 Kg.



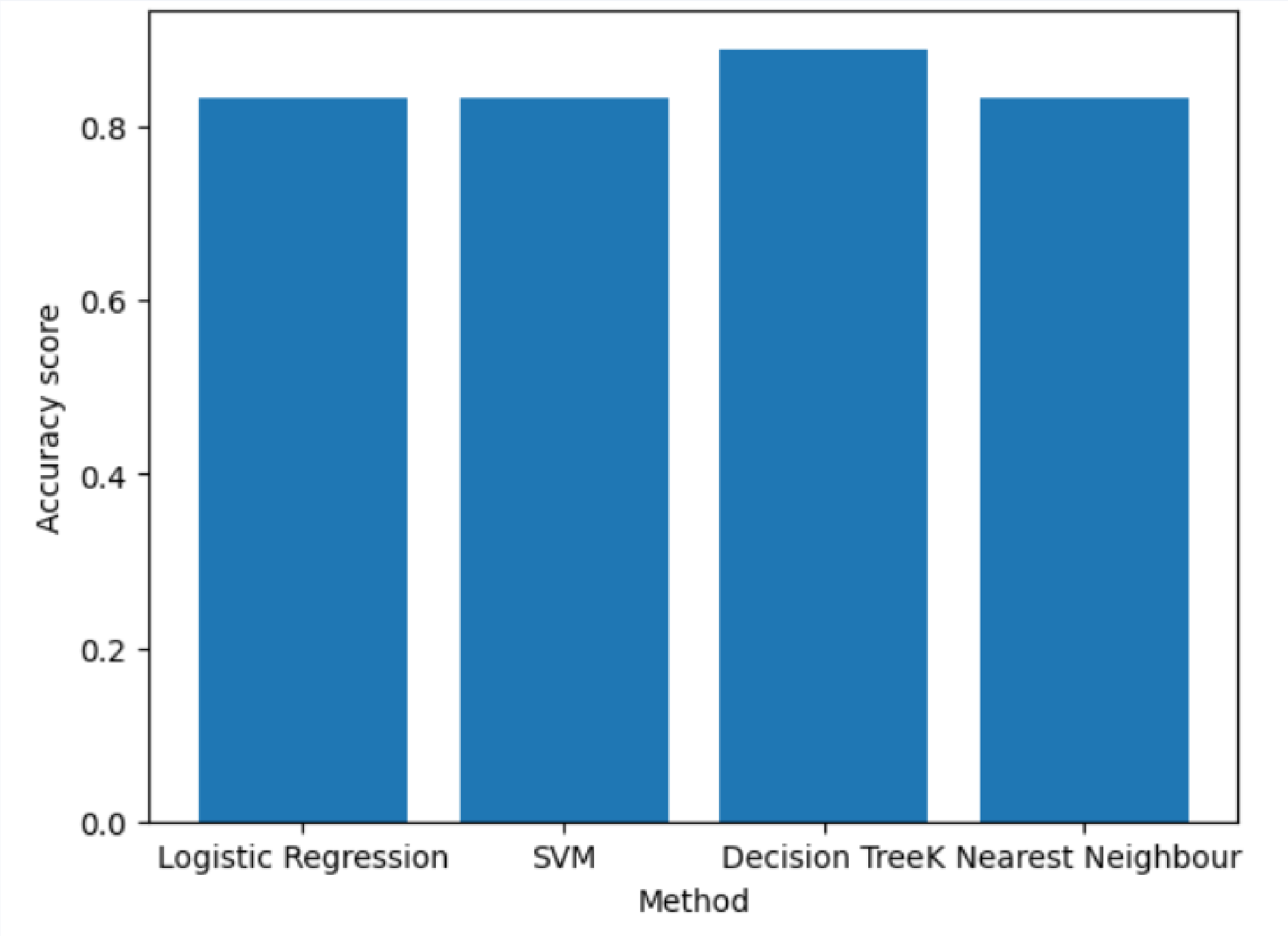
For this payload range only booster versions FT and B4 featured, and most launches were unsuccessful.

Section 5

Predictive Analysis (Classification)

Classification Accuracy

The model with the highest classification accuracy is the Decision Tree model with an accuracy percentage of 88.89%

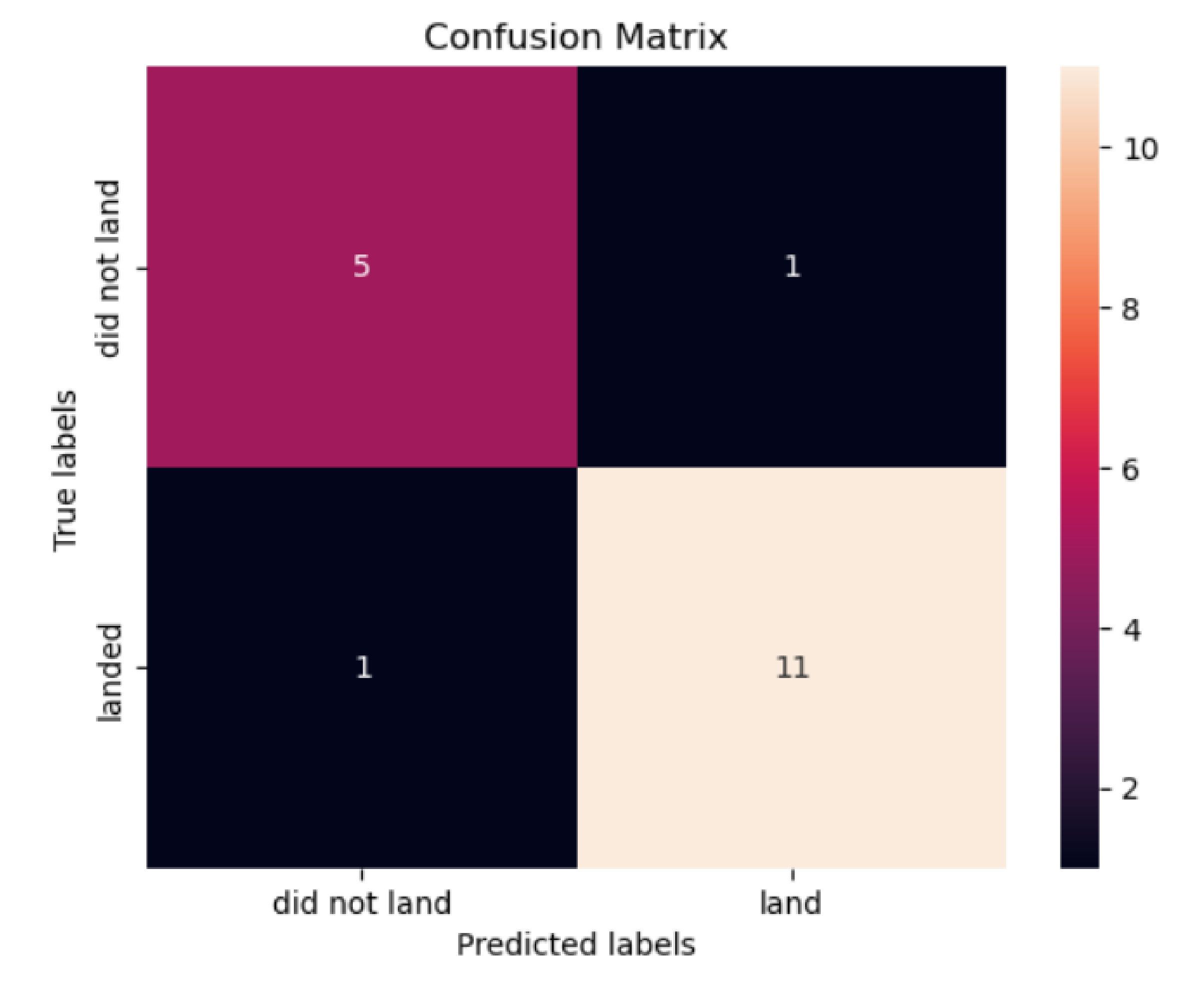


Confusion Matrix

The Decision Tree model performed the best as it had the highest accuracy score.

The confusion matrix for the model gave:

- 11 True Positives
- 5 True Negatives
- 1 False Positive
- 1 False Negative



Conclusions

- The number of successful launches increased as the number of flights increased, indicating a positive relation between the two.
- High Earth orbits such as ES-L1, GEO, HEO and SSO all had high success rates. Indicating that success rate is dependent on orbit type.
- The number of successes increased over time.
- All launch sites are near to coastlines, highways or railways, while being relatively far away from major cities.
- The KSC launch site has the highest number of successful landings.
- The Decision Tree Classifier performed the best as it had the highest accuracy score.
- All classification models performed well and had similar confusion matrices.

Appendix

- All notebooks used can be found on GitHub.

Thank you!